

左 聪

+61-88538287

zucong10@gmail.com

zucong10.github.io

Google Scholar: <https://scholar.google.com/citations?user=XvE1w8kAAAAJ&hl=en>

学习经历

- 2017.05~ 博士, 计算机科学与技术, 蒙纳士大学 (Monash University), 澳大利亚.
2020.11 ● 导师: 廖启瑞
● 副导师: 孙士锋, Josef Pieprzyk
- 2013.09~ 硕士, 计算机科学与技术, 浙江工商大学, 杭州, 浙江, 中国.
2016.06 ● 绩点: 86/100
● 导师: 邵俊, 凌云
- 2009.09~ 学士, 计算机科学与技术, 南京工程学院, 南京, 江苏, 中国.
2013.06 ● 绩点: 83/100
● 班主任: 柳圣

研究方向

对称可搜索加密, 数据安全与隐私保护, 应用密码学.

工作经历

- 2021.01~ 博士后, 南洋理工大学.
至今
- 2019.02~ 助教, 计算机系, 蒙纳士大学.
2020.06 ● 主要负责带软件安全实验课 (FIT 3173) 并改进其上课材料, 同时负责该课程的答疑课 (2019 第一学期, 2020 第一学期)

学术服务工作

受邀审稿人.

- The Computer Journal
- IEEE Transactions on Information Forensics and Security (TIFS)
- IEEE Transactions on Dependable and Secure Computing (TDSC)
- IEEE Transactions on Services Computing (TSC)
- The 25th Australasian Conference on Information Security and Privacy (ACISP 2020)
- The 21st International Conference on Information and Communications Security (ICICS 2019)

学术会议分组报告主持人.

- The 13th International Conference on Provable and Practical Security (PROVSEC 2019)
- The 13th International Conference on Network and System Security (NSS 2019)
- The 18th Annual International Conference on Privacy, Security and Trust (PST2021)

学术会议学术委员会成员.

- The 2022 IEEE International Conference on Communications (ICC): SAC E-Health Track
- The 18th Annual International Conference on Privacy, Security and Trust (PST2021)
- The 15th International Conference on Provable and Practical Security (PROVSEC 2021)
- The 17th EAI International Conference on Heterogeneous Networking for Quality, Reliability, Security and Robustness (EAI QSHINE 2021)
- The 13th EAI International Conference on Ad Hoc Networks (EAI ADHOCNETS 2021)

助理特约编辑.

- Special issue in MDPI Cryptography: Emerging Cryptographic Protocols for Blockchain and Its Applications

发表论文

- [1] **Cong Zuo**, Shi-Feng Sun, Joseph K. Liu, Jun Shao, Josef Pieprzyk, and Lei Xu. Forward and backward private dsse for range queries. In *IEEE Transactions on Dependable and Secure Computing (TDSC)*, accepted for publication, (**CCF A**, **IF=6.404**).
- [2] Shi-Feng Sun, **Cong Zuo**, Joseph K. Liu, Amin Sakzad, Ron Steinfeld, Tsz Hon Yuen, Xingliang Yuan, and Dawu Gu. Non-interactive multi-client searchable encryption: Realization and implementation. *IEEE Transactions on Dependable and Secure Computing (TDSC)*, accepted for publication, (**CCF A**, **IF=6.404**).
- [3] Shabnam Kasra Kermanshahi, Joseph K. Liu, Ron Steinfeld, Surya Nepal, Shangqi Lai, Randolph Loh, and **Cong Zuo**. Multi-client cloud-based symmetric searchable encryption. *IEEE Transactions on Dependable and Secure Computing (TDSC)*, accepted for publication, (**CCF A**, **IF=6.404**).
- [4] Lei Xu, Shi-Feng Sun, Xingliang Yuan, Joseph K. Liu, **Cong Zuo**, and Chungeng Xu. Enabling authorized encrypted search for multi-authority medical databases. *IEEE Transactions on Emerging Topics in Computing (TETC)*, accepted for publication, (**SCI Q2**, **IF=4.989**).
- [5] Lei Xu, Chungeng Xu, Joseph K. Liu, **Cong Zuo**, and Peng Zhang. Building a dynamic searchable encrypted medical database for multi-client. *Information Sciences*, 527:394–405, 2020, (**CCF B**, **SCI Q1**, **IF=5.910**).
- [6] Qingqing Gan, **Cong Zuo**, Jianfeng Wang, Shi-Feng Sun, and Xiaoming Wang. Dynamic searchable symmetric encryption with forward and backward privacy: A survey. In *Network and System Security (NSS)*, pages 37–52, Cham, 2019.
- [7] **Cong Zuo**, Shi-Feng Sun, Joseph K. Liu, Jun Shao, and Josef Pieprzyk. Dynamic searchable symmetric encryption with forward and stronger backward privacy. In *Computer Security - 24th European Symposium on Research in Computer Security (ESORICS)*, pages 283–303, 2019, (**CCF B**).

- [8] Randolph Loh, **Cong Zuo**, Joseph K. Liu, and Shi-Feng Sun. A multi-client dsse scheme supporting range queries. In *International Conference on Information Security and Cryptology (Inscrypt)*, pages 289–307, 2018.
- [9] Zhimei Sui, Shangqi Lai, **Cong Zuo**, Xingliang Yuan, Joseph K. Liu, and Haifeng Qian. An encrypted database with enforced access control and blockchain validation. In *International Conference on Information Security and Cryptology (Inscrypt)*, pages 260–273, 2018.
- [10] Shangqi Lai, Sikhar Patranabis, Amin Sakzad, Joseph K. Liu, Debdeep Mukhopadhyay, Ron Steinfeld, Shifeng Sun, Dongxi Liu, and **Cong Zuo**. Result pattern hiding searchable encryption for conjunctive queries. In *Proceedings of the 2018 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, pages 745–762, 2018, (**CCF A**).
- [11] **Cong Zuo**, Shifeng Sun, Joseph K. Liu, Jun Shao, and Josef Pieprzyk. Dynamic searchable symmetric encryption schemes supporting range queries with forward (and backward) security. In *Computer Security - 23rd European Symposium on Research in Computer Security (ESORICS)*, pages 228–246, 2018, (**CCF B**).
- [12] **Cong Zuo**, Jun Shao, Joseph K. Liu, Guiyi Wei, and Yun Ling. Fine-grained two-factor protection mechanism for data sharing in cloud storage. *IEEE Trans. Information Forensics and Security (TIFS)*, 13(1):186–196, 2018, (**CCF A, IF=6.211**).
- [13] **Cong Zuo**, Jun Shao, Guiyi Wei, Mande Xie, and Min Ji. Cca-secure ABE with outsourced decryption for fog computing. *Future Generation Computer Systems (FGCS)*, 78:730–738, 2018, (**CCF C, SCI Q2, ESI, IF=5.768**).
- [14] Xinxin Ma, Jun Shao, **Cong Zuo**, and Ru Meng. Efficient certificate-based signature and its aggregation. In *Information Security Practice and Experience - 13th International Conference (ISPEC)*, pages 391–408, 2017.
- [15] Lei Xu, Chungeng Xu, Joseph K. Liu, **Cong Zuo**, and Peng Zhang. A multi-client dynamic searchable symmetric encryption system with physical deletion. In *Information and Communications Security - 19th International Conference (ICICS)*, pages 516–528, 2017, (**CCF C**).
- [16] **Cong Zuo**, Kaitai Liang, Zoe L. Jiang, Jun Shao, and Jun-bin Fang. Cost-effective privacy-preserving vehicular urban sensing system. *Personal and Ubiquitous Computing*, 21(5):893–901, 2017, (**IF=2.0**).
- [17] **Cong Zuo**, Jun Shao, Zhe Liu, Yun Ling, and Guiyi Wei. Hidden-token searchable public-key encryption. In *2017 IEEE Trustcom/BigDataSE/ICSS, Sydney, Australia, August 1-4, 2017*, pages 248–254, 2017, (**CCF C**).
- [18] **Cong Zuo**, James Macindoe, Siyin Yang, Ron Steinfeld, and Joseph K. Liu. Trusted boolean search on cloud using searchable symmetric encryption. In *2016 IEEE Trustcom/BigDataSE/ISPA*, pages 113–120, 2016, (**CCF C**).

- [19] **Cong Zuo**, Jun Shao, Guiyi Wei, Mande Xie, and Min Ji. Chosen ciphertext secure attribute-based encryption with outsourced decryption. In *Information Security and Privacy - 21st Australasian Conference (ACISP)*, pages 495–508, 2016, (**CCF C**).
- [20] Jun Shao, Xiaodong Lin, Rongxing Lu, and **Cong Zuo**. A threshold anonymous authentication protocol for vanets. *IEEE Transactions on Vehicular Technology*, 65(3):1711–1720, 2016, (**SCI Q2, IF=5.339**).
- [21] Jun Shao, Rongxing Lu, Xiaodong Lin, and **Cong Zuo**. New threshold anonymous authentication for vanets. In *2015 IEEE/CIC International Conference on Communications in China*, pages 1–6, 2015.

获奖经历

- 2017~2020 **Data61 博士和补充奖学金**, Data61, 澳大利亚联邦科学与工业研究组织 (CSIRO).
- 2017~2020 **国际研究生研究奖学金**, 蒙纳士大学.
- 2016 **王光明奖学金**, 浙江工商大学.
- 2013 **优秀毕业生**, 南京工程学院.
- 2012 **校级三好学生**, 南京工程学院.

学生指导

- 2018 **Randolph Loh**, 蒙纳士大学硕士研究生. 在他读研期间, 参与指导其研究, 其研究结果发表在 *International Conference on Information Security and Cryptology (Inscrypt)*, 2018 (见论文 [8]).
- 2018 **隋智美**, 华东师范大学硕士研究生, 并于 2018 年访问蒙纳士大学. 在她访问期间, 参与指导其研究, 其研究成果发表在 *International Conference on Information Security and Cryptology (Inscrypt)*, 2018 (见论文 [9]).

推荐人

廖启瑞, 蒙纳士大学, 澳大利亚, joseph.liu@monash.edu.

邵俊, 浙江工商大学, 中国, chn.junshao@gmail.com.

Josef Pieprzyk, Data61 高级首席科研人员, CSIRO, 澳大利亚, Josef.Pieprzyk@data61.csiro.au.



Cong Zuo

Nanyang Technological University
Cryptography

GET MY OWN PROFILE

	All	Since 2017
Citations	702	683
h-index	10	10
i10-index	11	11

13 articles

7 articles

not available

available

Based on funding mandates

TITLE	CITED BY	YEAR
A threshold anonymous authentication protocol for VANETs J Shao, X Lin, R Lu, C Zuo IEEE Transactions on vehicular technology 65 (3), 1711-1720	220	2015
CCA-secure ABE with outsourced decryption for fog computing C Zuo, J Shao, G Wei, M Xie, M Ji Future Generation Computer Systems 78, 730-738	123	2018
Result pattern hiding searchable encryption for conjunctive queries S Lai, S Patranabis, A Sakzad, JK Liu, D Mukhopadhyay, R Steinfeld, ... Proceedings of the 2018 ACM SIGSAC Conference on Computer and Communications ...	76	2018
Fine-grained two-factor protection mechanism for data sharing in cloud storage C Zuo, J Shao, JK Liu, G Wei, Y Ling IEEE Transactions on Information Forensics and Security 13 (1), 186-196	50	2017
Dynamic searchable symmetric encryption schemes supporting range queries with forward (and backward) security C Zuo, SF Sun, JK Liu, J Shao, J Pieprzyk European Symposium on Research in Computer Security, 228-246	43	2018
Privacy-preserving COVID-19 contact tracing app: a zero-knowledge proof approach JK Liu, MH Au, TH Yuen, C Zuo, J Wang, A Sakzad, X Luo, L Li, ... Cryptology ePrint Archive	39	2020
Dynamic searchable symmetric encryption with forward and stronger backward privacy C Zuo, SF Sun, JK Liu, J Shao, J Pieprzyk European Symposium on Research in Computer Security, 283-303	34	2019
Trusted boolean search on cloud using searchable symmetric encryption C Zuo, J Macindoe, S Yang, R Steinfeld, JK Liu 2016 IEEE Trustcom/BigDataSE/ISPA, 113-120	27	2016
Building a dynamic searchable encrypted medical database for multi-client L Xu, C Xu, JK Liu, C Zuo, P Zhang Information Sciences 527, 394-405	14	2020
New threshold anonymous authentication for VANETs J Shao, R Lu, X Lin, C Zuo 2015 IEEE/CIC International Conference on Communications in China (ICCC), 1-6	12	2015

TITLE	CITED BY	YEAR
Forward and backward private dsse for range queries C Zuo, S Sun, JK Liu, J Shao, J Pieprzyk, L Xu IEEE Transactions on Dependable and Secure Computing	10	2020
Enabling authorized encrypted search for multi-authority medical databases L Xu, S Sun, X Yuan, JK Liu, C Zuo, C Xu IEEE Transactions on Emerging Topics in Computing	7	2019
Cost-effective privacy-preserving vehicular urban sensing system C Zuo, K Liang, ZL Jiang, J Shao, J Fang Personal and Ubiquitous Computing 21 (5), 893-901	7	2017
Efficient certificate-based signature and its aggregation X Ma, J Shao, C Zuo, R Meng International Conference on Information Security Practice and Experience ...	6	2017
Multi-client cloud-based symmetric searchable encryption SK Kermanshahi, JK Liu, R Steinfeld, S Nepal, S Lai, R Loh, C Zuo IEEE Transactions on Dependable and Secure Computing	5	2019
Chosen ciphertext secure attribute-based encryption with outsourced decryption C Zuo, J Shao, G Wei, M Xie, M Ji Australasian Conference on Information Security and Privacy, 495-508	5	2016
Dynamic searchable symmetric encryption schemes supporting range queries with forward/backward privacy C Zuo, SF Sun, JK Liu, J Shao, J Pieprzyk arXiv preprint arXiv:1905.08561	4	2019
A multi-client DSSE scheme supporting range queries R Loh, C Zuo, JK Liu, SF Sun International Conference on Information Security and Cryptology, 289-307	4	2018
Non-Interactive Multi-Client Searchable Encryption: Realization and Implementation SF Sun, C Zuo, JK Liu, A Sakzad, R Steinfeld, TH Yuen, X Yuan, D Gu IEEE Transactions on Dependable and Secure Computing	3	2020
An encrypted database with enforced access control and blockchain validation Z Sui, S Lai, C Zuo, X Yuan, JK Liu, H Qian International Conference on Information Security and Cryptology, 260-273	3	2018
hpress: A hardware-enhanced proxy re-encryption scheme using secure enclave F Zhang, Z Liang, C Zuo, J Shao, J Ning, J Sun, JK Liu, Y Bao IEEE Transactions on Computer-Aided Design of Integrated Circuits and ...	2	2020
Forward and backward private dynamic searchable symmetric encryption for conjunctive queries C Zuo, SF Sun, JK Liu, J Shao, J Pieprzyk, G Wei Cryptology ePrint Archive	2	2020

TITLE	CITED BY	YEAR
A multi-client dynamic searchable symmetric encryption system with physical deletion L Xu, C Xu, JK Liu, C Zuo, P Zhang International Conference on Information and Communications Security, 516-528	2	2017
Hidden-Token Searchable Public-Key Encryption C Zuo, J Shao, Z Liu, Y Ling, G Wei 2017 IEEE Trustcom/BigDataSE/ICCESS, 248-254	2	2017
Searchable Encryption with Access Control on Keywords in Multi-User Setting L Li, C Xu, X Yu, B Dou, C Zuo Journal of Cybersecurity 2 (1), 9	1	2020
Dynamic Searchable Symmetric Encryption with Forward and Backward Privacy: A Survey Q Gan, C Zuo, J Wang, SF Sun, X Wang International Conference on Network and System Security, 37-52	1	2019
Privacy-Preserving Contact Tracing Protocol for Mobile Devices: A Zero-Knowledge Proof Approach JK Liu, MH Au, TH Yuen, C Zuo, J Wang, A Sakzad, X Luo, L Li, ... International Conference on Information Security Practice and Experience ...		2021
IoT Services: Realizing Private Real-Time Detection via Authenticated Conjunctive Searchable Encryption C Xu, M Lin, J Cheng, Y Zhao, C Zuo Journal of Cybersecurity 3 (1), 55		2021
Searchable Encryption for Conjunctive Queries with Extended Forward and Backward Privacy C Zuo, S Lai, X Yuan, JK Liu, J Shao, H Wang Cryptology ePrint Archive		2021
Verifiable Identity-Based Encryption with Keyword Search for IoT from Lattice L Mei, C Xu, L Xu, X Yu, C Zuo CMC-COMPUTERS MATERIALS & CONTINUA 68 (2), 2299-2314		2021
Attacking the Niederreiter-type cryptosystem based on rank metric C Xu, Y Zhang, L Mei, L Xu, C Zuo International Journal of Embedded Systems 13 (4), 398-404		2020
Enhanced Security for Searchable Symmetric Encryption Supporting Rich Queries C ZUO Monash University		